

UNIT - III

5

MULTIPLE INTEGRALS

* VOLUME INTEGRALS :

- 1) Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ by using triple integral.

Given:

$$\begin{array}{l|l|l} x^2 + y^2 + z^2 = a^2 & \text{put } z=0 & \text{put } z=0 \text{ \& } y=0 \\ z^2 = a^2 - x^2 - y^2 & x^2 + y^2 = a^2 & x^2 = a^2 \\ y^2 = a^2 - x^2 & y^2 = a^2 - x^2 & x = \pm a \\ z = \pm \sqrt{a^2 - x^2 - y^2} & y = \pm \sqrt{a^2 - x^2} & \end{array}$$

$$dv = dx dy dz \text{ (or) } dz dy dx$$

$$\text{Volume} = 8 \iiint dv$$

$$= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz dy dx$$

$$= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} [z]_0^{\sqrt{a^2-x^2-y^2}} dy dx$$

$$= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{(a^2-x^2)-y^2} dy dx$$

$$= 8 \int_0^a \left[\frac{y}{2} \sqrt{a^2-x^2-y^2} + \frac{a^2-x^2}{2} \sin^{-1} \left(\frac{y}{\sqrt{a^2-x^2}} \right) \right] dx$$

$$= 8 \int_0^a \left\{ 0 + \frac{a^2-x^2}{2} \sin^{-1} \left(\frac{\sqrt{a^2-x^2}}{\sqrt{a^2-x^2}} \right) \right\} - \{0+0\} dx$$

$$= 8 \int_0^a \frac{a^2-x^2}{2} \sin^{-1}(1) dx$$

$$= \frac{8}{2} \int_0^a (a^2-x^2) \frac{\pi}{2} dx$$

$$= 2\pi \int_0^a (a^2-x^2) dx$$

$$= 2\pi \left[a^2x - \frac{x^3}{3} \right]_0^a$$

$$= 2\pi \left[\left(a^3 - \frac{a^3}{3} \right) - (0-0) \right]$$

$$= 2\pi \times \frac{2a^3}{3}$$

$$= \frac{4\pi a^3}{3} \text{ cu. units}$$

$$\therefore \int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right)$$

2) Evaluate $\iiint_R xyz \, dx \, dy \, dz$ through the +ve octant of the sphere $x^2 + y^2 + z^2 \leq a^2$

3)

c

Find the +ve octant of the sphere $x^2 + y^2 + z^2 \leq a^2$

x varies from 0 to $\sqrt{a^2 - y^2 - z^2}$

y varies from 0 to $\sqrt{a^2 - x^2}$

$$dv = dx \, dy \, dz = dz \, dy \, dx$$

z varies from 0 to a

$$\begin{aligned} \iiint_R xyz \, dx \, dy \, dz &= \int_0^a \int_0^{\sqrt{a^2 - x^2}} \int_0^{\sqrt{a^2 - x^2 - y^2}} xyz \, dz \, dy \, dx \\ &= \int_0^a \int_0^{\sqrt{a^2 - x^2}} xy \left[\frac{z^2}{2} \right]_0^{\sqrt{a^2 - x^2 - y^2}} dy \, dx \\ &= \frac{1}{2} \int_0^a \int_0^{\sqrt{a^2 - x^2}} xy (a^2 - x^2 - y^2) dy \, dx \\ &= \frac{1}{2} \int_0^a \int_0^{\sqrt{a^2 - x^2}} [x(a^2 - x^2)y - xy^3] dy \, dx \\ &= \frac{1}{2} \int_0^a \left[x(a^2 - x^2) \frac{y^2}{2} - \frac{xy^4}{4} \right]_0^{\sqrt{a^2 - x^2}} dx \\ &= \frac{1}{2} \int_0^a \left[\frac{x(a^2 - x^2)(a^2 - x^2)}{2} - \frac{x(a^2 - x^2)^2}{4} \right] dx \\ &= \frac{1}{2} \times \frac{1}{4} \int_0^a [2x(a^2 - x^2)^2 - x(a^2 - x^2)^2] dx \\ &= \frac{1}{8} \int_0^a x(a^2 - x^2)^2 dx \\ &= \frac{1}{8} \int_0^a x [a^4 + x^4 - 2a^2x^2] dx \\ &= \frac{1}{8} \int_0^a (a^4x + x^5 - 2a^2x^3) dx \\ &= \frac{1}{8} \left[\frac{a^4x^2}{2} + \frac{x^6}{6} - \frac{2a^2x^4}{4} \right]_0^a \\ &= \frac{1}{8} \left[\frac{a^6}{2} + \frac{a^6}{6} - \frac{2a^6}{4} \right] \\ &= \frac{a^6}{48} \text{ cu. units} // \end{aligned}$$

(2)

3) Evaluate $\iiint \frac{dx dy dz}{(x+y+z+1)^3}$ over the region bounded by the coordinate planes and the plane $x+y+z=1$.

Since the region bounded by the co-ordinate planes $x=0$, $y=0$ & $z=0$ and the plane $x+y+z=1$

$$\begin{array}{c|c|c} x+y+z=1 & \text{put } z=0 & \text{put } z=0, y=0 \\ z=1-x-y & x+y=1 & x=1 \\ & y=1-x & \end{array}$$

z varies from 0 to $1-x-y$

y varies from 0 to $1-x$

x varies from 0 to 1

$$dv = dx dy dz = dz dy dx$$

$$\begin{aligned} \iiint_R \frac{dx dy dz}{(x+y+z+1)^3} &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dz dy dx}{(x+y+z+1)^3} \\ &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x+y+z+1)^{-3} dz dy dx \\ &= \int_0^1 \int_0^{1-x} \left[\frac{(x+y+z+1)^{-2}}{-2} \right]_0^{1-x-y} dy dx \\ &= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left\{ (x+y+1-x-y+1)^{-2} - (x+y+1)^{-2} \right\} dy dx \\ &= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left\{ (2)^{-2} - (x+y+1)^{-2} \right\} dy dx \\ &= -\frac{1}{2} \int_0^1 \left[\frac{1}{4} y - \frac{(x+y+1)^{-1}}{-1} \right]_0^{1-x} dx \\ &= -\frac{1}{2} \int_0^1 \left[\left\{ \frac{1-x}{4} + (x+1-x+1)^{-1} \right\} - \left\{ 0 + (x+1)^{-1} \right\} \right] dx \\ &= -\frac{1}{2} \int_0^1 \left[-\frac{1}{4} + \frac{x}{4} + \frac{1}{2} - \frac{1}{1+x} \right] dx \\ &= -\frac{1}{2} \left[\frac{x}{4} - \frac{x^2}{8} + \frac{x}{2} - \log(1+x) \right]_0^1 \\ &= -\frac{1}{2} \left\{ \left[\frac{1}{4} - \frac{1}{8} + \frac{1}{2} - \log 2 \right] - \left[0 - 0 + 0 - \log 1 \right] \right\} \\ &= -\frac{1}{2} \left\{ \frac{2-1+4}{8} - \log 2 \right\} \\ &= -\frac{1}{2} \left\{ +\frac{5}{8} - \log 2 \right\} \\ &= \frac{1}{2} \left[\log 2 - \frac{5}{8} \right] \quad \text{cu. units} \end{aligned}$$

4) Evaluate

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}}$$

$$\because \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)$$

6)

$$\begin{aligned} \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}} &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left\{ \int_0^{\sqrt{1-x^2-y^2}} \frac{dz}{\sqrt{1-x^2-y^2-z^2}} \right\} dy dx \\ &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1} \left(\frac{z}{\sqrt{1-x^2-y^2}} \right) \right]_0^{\sqrt{1-x^2-y^2}} dy dx \\ &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left\{ \sin^{-1}(1) - \sin^{-1}(0) \right\} dy dx \\ &= \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{\pi}{2} dy dx \\ &= \frac{\pi}{2} \int_0^1 [y]_0^{\sqrt{1-x^2}} dx \\ &= \frac{\pi}{2} \int_0^1 \sqrt{1-x^2} dx \\ &= \frac{\pi}{2} \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} \left(\frac{x}{1} \right) \right]_0^1 \\ &= \frac{\pi}{2} \left\{ \left[0 + \frac{1}{2} \sin^{-1}(1) \right] - \left[0 + 0 \right] \right\} \\ &= \frac{\pi}{4} \times \frac{\pi}{2} \\ &= \frac{\pi^2}{8} \end{aligned}$$

5) Evaluate:

$$\begin{aligned} \int_0^a \int_0^b \int_0^c (x+y+z) dx dy dz &= \int_0^a \int_0^b \left[\frac{x^2}{2} + (y+z)x \right]_0^c dy dz \\ &= \int_0^a \int_0^b \left[\frac{c^2}{2} + (y+z)c \right] dy dz \\ &= \int_0^a \int_0^b \left[\left(cz + \frac{c^2}{2} \right) + cy \right] dy dz \\ &= \int_0^a \left[\left(cz + \frac{c^2}{2} \right) y + \frac{cy^2}{2} \right]_0^b dz \\ &= \int_0^a \left[\left(cx + \frac{c^2}{2} \right) b + \frac{cb^2}{2} \right] dz \\ &= \int_0^a \left[bcx + \frac{bc^2}{2} + \frac{cb^2}{2} \right] dz \\ &= \left[\frac{bcx^2}{2} + \frac{bc^2x}{2} + \frac{cb^2x}{2} \right]_0^a \\ &= \frac{a^2bc}{2} + \frac{abc^2}{2} + \frac{ab^2c}{2} = \frac{abc}{2} (a+b+c) \end{aligned}$$

(4)

- b) Find the volume of the tetrahedron bounded by the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ and coordinate planes $x=0$, $y=0$ & $z=0$

Since the region bounded by the co-ordinate plane $x=0$, $y=0$ & $z=0$ and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\begin{array}{l} \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \\ \frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b} \\ z = c \left(1 - \frac{x}{a} - \frac{y}{b} \right) \end{array} \quad \left| \begin{array}{l} \text{put } z=0 \\ \frac{x}{a} + \frac{y}{b} = 1 \\ \frac{y}{b} = 1 - \frac{x}{a} \\ y = b \left(1 - \frac{x}{a} \right) \end{array} \right. \quad \left| \begin{array}{l} \text{put } x=0, y=0 \\ \frac{z}{c} = 1 \\ z = c \end{array} \right.$$

x varies from 0 to a

y varies from 0 to $b \left(1 - \frac{x}{a} \right)$

z varies from 0 to $c \left(1 - \frac{x}{a} - \frac{y}{b} \right)$

$$dv = dx dy dz \quad \therefore \quad dz dy dx$$

$$\begin{aligned} \text{volume} &= \iiint_R dz dy dx \\ &= \int_0^a \int_0^{b(1-\frac{x}{a})} \int_0^{c(1-\frac{x}{a}-\frac{y}{b})} dz dy dx \\ &= \int_0^a \int_0^{b(1-\frac{x}{a})} \left[z \right]_0^{c(1-\frac{x}{a}-\frac{y}{b})} dy dx \\ &= \int_0^a \int_0^{b(1-\frac{x}{a})} c \left(1 - \frac{x}{a} - \frac{y}{b} \right) dy dx \\ &= c \int_0^a \int_0^{b(1-\frac{x}{a})} \left[\left(1 - \frac{x}{a} \right) y - \frac{y^2}{2b} \right] dy dx \\ &= c \int_0^a \left[\left(1 - \frac{x}{a} \right) y - \frac{y^2}{2b} \right]_0^{b(1-\frac{x}{a})} dx \\ &= c \int_0^a \left[b \left(1 - \frac{x}{a} \right)^2 - \frac{b^2 \left(1 - \frac{x}{a} \right)^2}{2b} \right] dx \\ &= \frac{cb}{2} \int_0^a \left(1 - \frac{x}{a} \right)^2 dx \\ &= \frac{bc}{2} \int_0^a \left(1 + \frac{x^2}{a^2} - \frac{2x}{a} \right) dx \\ &= \frac{bc}{2} \left[x + \frac{x^3}{3a^2} - \frac{2x^2}{2a} \right]_0^a \end{aligned}$$

$$= \frac{bc}{2} \left[\left\{ a + \frac{a^3}{3a^2} - \frac{a^2}{a} \right\} - 0 \right]$$

$$= \frac{abc}{6} \text{ cu. units.}$$

7) Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Given:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

$$z^2 = c^2 \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right)$$

$$z = \pm c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$$

put $z=0$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left(1 - \frac{x^2}{a^2} \right)$$

$$y = \pm b \sqrt{1 - \frac{x^2}{a^2}}$$

put $z=0, y=0$

$$\frac{x^2}{a^2} = 1$$

$$x^2 = a^2$$

$$x = \pm a$$

8)

x varies from $-a$ to a

y varies from $-b\sqrt{1 - \frac{x^2}{a^2}}$ to $b\sqrt{1 - \frac{x^2}{a^2}}$

z varies from $-c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$ to $c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$

$$\text{Volume} = 8 \int_0^a \int_0^{b\sqrt{1 - \frac{x^2}{a^2}}} \int_0^{c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} dz dy dx$$

$$= 8 \int_0^a \int_0^{b\sqrt{1 - \frac{x^2}{a^2}}} \left[z \right]_0^{c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}} dy dx$$

$$= 8 \int_0^a \int_0^{b\sqrt{1 - \frac{x^2}{a^2}}} c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} dy dx$$

$$= 8c \int_0^a \int_0^{b\sqrt{1 - \frac{x^2}{a^2}}} \frac{1}{b} \sqrt{b^2 \left(1 - \frac{x^2}{a^2} \right) - y^2} dy dx$$

$$= \frac{8c}{b} \int_0^a \left[\frac{y}{2} \sqrt{b^2 \left(1 - \frac{x^2}{a^2} \right) - y^2} + \frac{b^2 \left(1 - \frac{x^2}{a^2} \right)}{2} \sin^{-1} \left(\frac{y}{b\sqrt{1 - \frac{x^2}{a^2}}} \right) \right]_0^{b\sqrt{1 - \frac{x^2}{a^2}}} dx$$

$$= \frac{8c}{2b} \int_0^a \left\{ b^2 \left(1 - \frac{x^2}{a^2} \right) \sin^{-1}(1) - 0 \right\} dx$$

$$= \frac{4c}{2b} \cdot b^2 \int_0^a \left(1 - \frac{x^2}{a^2} \right) \pi_2 dx$$

$$= 2bc\pi \left[x - \frac{x^3}{3a^2} \right]_0^a$$

(6)

$$= 2bc\pi \left[a - \frac{a^3}{3a^2} \right]$$

$$= 2bc\pi \left[\frac{3a - a}{3} \right]$$

$$= 2bc\pi \times \frac{2a}{3}$$

$$= \frac{4abc\pi}{3} \text{ cu. units.}$$

8) Evaluate :

$$\begin{aligned} \int_0^1 \int_0^{1-x} \int_0^{1-x-y} xyz \, dx \, dy \, dz &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} xyz \, dx \, dy \, dz \\ &= \int_0^1 \int_0^{1-x} xy \left[\frac{z^2}{2} \right]_0^{1-x-y} dy \, dx \\ &= \frac{1}{2} \int_0^1 \int_0^{1-x} xy (1-x-y)^2 dy \, dx \\ &= \frac{1}{2} \int_0^1 \int_0^{1-x} xy [(1-x)^2 + y^2 - 2y(1-x)] dy \, dx \\ &= \frac{1}{2} \int_0^1 \left[x(1-x)^2 \frac{y^2}{2} + x \frac{y^4}{4} - 2x(1-x) \frac{y^3}{3} \right]_0^{1-x} dx \\ &= \frac{1}{2} \int_0^1 \left[\frac{x(1-x)^4}{2} + \frac{x(1-x)^4}{4} - 2x(1-x)^4 \cdot \frac{2}{3} \right] dx \\ &= \frac{1}{2} \int_0^1 x(1-x)^4 \left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right] dx \\ &= \frac{1}{2} \int_0^1 x(1-x)^4 \cdot \frac{1}{12} dx \\ &= \frac{1}{24} \int_0^1 x(1-x)^4 dx \\ &= \frac{1}{24} \int_0^1 x(1 - 4x + 6x^2 - 4x^3 + x^4) dx \\ &= \frac{1}{24} \int_0^1 (x - 4x^2 + 6x^3 - 4x^4 + x^5) dx \\ &= \frac{1}{24} \left[\frac{x^2}{2} - \frac{4x^3}{3} + \frac{6x^4}{4} - \frac{4x^5}{5} + \frac{x^6}{6} \right]_0^1 \\ &= \frac{1}{24} \left[\frac{1}{2} - \frac{4}{3} + \frac{3}{2} - \frac{4}{5} + \frac{1}{6} \right] \\ &= \frac{1}{24} \times \frac{1}{30} = \frac{1}{720} \end{aligned}$$

9) Find the value of $\iiint dx dy dz$ through the positive spherical octant for which $x^2 + y^2 + z^2 \leq a^2$

Assume that the spherical co-ordinates is

$$x = r \sin \theta \cos \phi ; y = r \sin \theta \sin \phi ; z = r \cos \theta$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

r varies from 0 to a

θ varies from 0 to π

ϕ varies from 0 to 2π

$$\begin{aligned} \iiint dx dy dz &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^a r^2 \sin \theta dr d\theta d\phi \\ &= \int_{\phi=0}^{2\pi} d\phi \cdot \int_{\theta=0}^{\pi} \sin \theta d\theta \cdot \int_{r=0}^a r^2 dr \\ &= [\phi]_0^{2\pi} \cdot [-\cos \theta]_0^{\pi} \cdot \left[\frac{r^3}{3}\right]_0^a \\ &= 2\pi \cdot (-\cos \pi + \cos 0) \cdot \frac{a^3}{3} \\ &= 2\pi \cdot (1+1) \cdot \frac{a^3}{3} \\ &= \frac{4\pi a^3}{3} \text{ cu. units} \end{aligned}$$

10) Evaluate $\iiint_V xyz dx dy dz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$.

Transform to cylindrical co-ordinates

$$x = r \cos \theta ; y = r \sin \theta ; z = z$$

$$dv = dx dy dz = r dr d\theta dz$$

The sphere becomes $r^2 + z^2 = a^2$

$$\Rightarrow z^2 = a^2 - r^2$$

$$z = \pm \sqrt{a^2 - r^2}$$

$$r \rightarrow 0 \text{ to } a$$

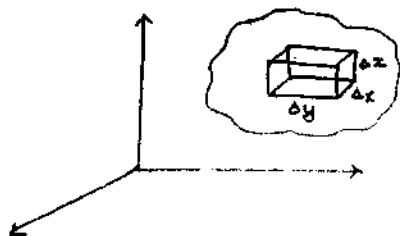
$$\theta \rightarrow 0 \text{ to } \frac{\pi}{2}$$

(Ist octant)

$$\begin{aligned} \iiint_V xyz dx dy dz &= \iiint_V xyz dv \\ &= \int_{\theta=0}^{\frac{\pi}{2}} \int_{r=0}^a \int_{z=0}^{\sqrt{a^2-r^2}} r \cos \theta \cdot r \sin \theta \cdot z r dz dr d\theta \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\pi/2} \int_0^a r^3 \cos \theta \sin \theta \left[\frac{z^2}{2} \right]_0^{\sqrt{a^2-r^2}} dr d\theta \\
&= \frac{1}{2} \int_0^{\pi/2} \int_0^a r^3 \cos \theta \sin \theta (a^2 - r^2) dr d\theta \\
&= \frac{1}{2} \int_0^{\pi/2} \int_0^a \cos \theta \sin \theta (a^2 r^3 - r^5) dr d\theta \\
&= \frac{1}{2} \int_0^{\pi/2} \cos \theta \sin \theta \left[\frac{a^2 r^4}{4} - \frac{r^6}{6} \right]_0^a d\theta \\
&= \frac{1}{2} \int_0^{\pi/2} \frac{\sin 2\theta}{2} \left[\frac{3a^6}{12} - \frac{2a^6}{12} \right] d\theta \\
&= \frac{a^6}{24} \int_0^{\pi/2} \frac{\sin 2\theta}{2} d\theta \\
&= \frac{a^6}{24} \left[-\frac{\cos 2\theta}{4} \right]_0^{\pi/2} \\
&= \frac{a^6}{24} \left[-\frac{\cos \pi}{4} + \frac{\cos 0}{4} \right] \\
&= \frac{a^6}{24} \left[\frac{2}{4} \right] \\
&= \frac{a^6}{48} \text{ cu. units.}
\end{aligned}$$

Volume by Triple Integration:



Divide the solid by planes parallel to the co-ordinate planes into rectangular parallelepipeds with dimensions Δx , Δy and Δz

Volume of parallelepiped = $\Delta x \Delta y \Delta z$

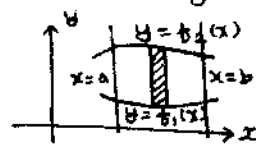
Total volume = $\lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0 \\ \Delta z \rightarrow 0}} \sum \Delta x \Delta y \Delta z = \iiint_R dx dy dz$

where R stands for the region of integration.

* Area by double integration:

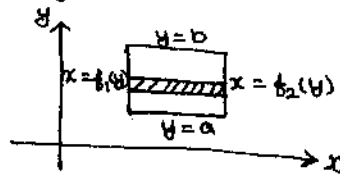
- 1) The area enclosed by the two curves $y = f_1(x)$ and $y = f_2(x)$ between the ordinates $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b \int_{f_1(x)}^{f_2(x)} dy dx$$



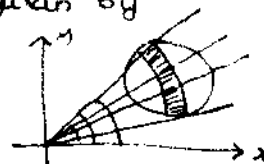
- 2) The area enclosed by the two curves $x = f_1(y)$ and $x = f_2(y)$ between the ordinates $y = a$ and $y = b$ is given by

$$\text{Area} = \int_{y=a}^b \int_{x=f_1(y)}^{x=f_2(y)} dx dy$$



- 3) The area enclosed by the curve $r = f(\theta)$ and between the ordinates $\theta = \alpha$ and $\theta = \beta$ is given by

$$\text{Area} = \int_{\theta=\alpha}^{\theta=\beta} \int_{r=f_1(\theta)}^{r=f_2(\theta)} r dr d\theta$$



- 1) Find the area between the parabolas $y^2 = 9x$ and $x^2 = 9y$

Given: $y^2 = 9x$ — (1)

$x^2 = 9y \Rightarrow y = \frac{x^2}{9}$ — (2)

Sub (2) in eqn (1)

$$\left(\frac{x^2}{9}\right)^2 = 9x$$

$$\frac{x^4}{9^2} = 9x$$

$$x^4 = 9^3 x$$

$$x(x^3 - 9^3) = 0$$

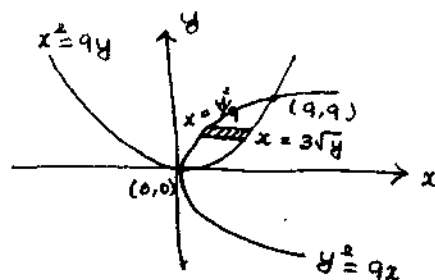
$$x(x-9)(x^2+9x+81) = 0$$

$$x = 0 \mid x - 9 = 0 \mid x^2 + 9x + 81 = 0$$

unreal roots are impossible

Sub $x=0$; $x=9$ in eqn (2) we get $y=0$; $y=9$

The points of intersection are $(0,0)$ and $(9,9)$



Since x varies from $\frac{y^2}{9}$ to $3\sqrt{y}$

y varies from 0 to 9

$$\text{Area} = \int_0^9 \int_{y^2/9}^{3\sqrt{y}} dx dy$$

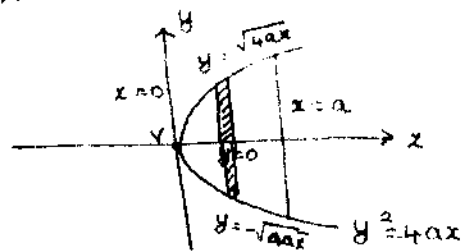
$$= \int_0^9 [x]_{y^2/9}^{3\sqrt{y}} dy$$

$$\begin{aligned}
 &= \int_0^9 \left[3\sqrt{y} - \frac{y^2}{9} \right] dy \\
 &= \int_0^9 \left[\frac{3y^{3/2}}{3/2} - \frac{y^3}{27} \right] dy \\
 &= 2(9)^{3/2} - \frac{9^3}{27} \\
 &= 2 \times 9 \times 3 - 27 \\
 &= 54 - 27 \\
 &= 27 \text{ sq. units}
 \end{aligned}$$

* Show that the area enclosed between the parabola $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$.

* Find the area between $y^2 = x$ and $x^2 = y$ by double integration.

2) Find the area of the parabola $y^2 = 4ax$ cut off by its latus rectum.



Given:

Equation of the parabola is $y^2 = 4ax$ and latus rectum $x = a$

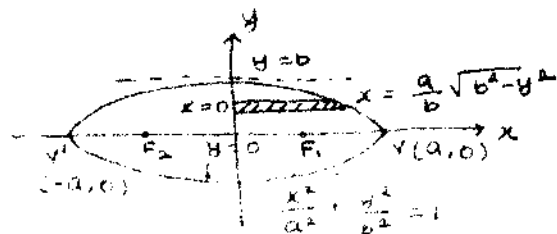
Limits: x varies from 0 to a

y varies from $-2\sqrt{ax}$ to $2\sqrt{ax}$

$$\begin{aligned}
 \text{Area} &= \int_0^a \int_{-2\sqrt{ax}}^{2\sqrt{ax}} dy dx \\
 &= 2 \int_0^a \int_0^{2\sqrt{ax}} dy dx \\
 &= 2 \int_0^a [y]_0^{2\sqrt{ax}} dx \\
 &= 2 \int_0^a 2\sqrt{ax} dx \\
 &= 4\sqrt{a} \int_0^a \sqrt{x} dx \\
 &= 4\sqrt{a} \left(\frac{x^{3/2}}{3/2} \right)_0^a \\
 &= \frac{8}{3} \sqrt{a} \cdot a\sqrt{a} \\
 &= \frac{8}{3} a^2 \text{ sq. units}
 \end{aligned}$$

3) Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Area



Given: Equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2}$$

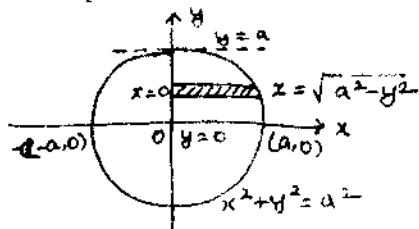
$$x^2 = \frac{a^2}{b^2} (b^2 - y^2)$$

$$x = \pm \frac{a}{b} \sqrt{b^2 - y^2}$$

Area = 4 × Area of the ellipse in I quadrant

$$\begin{aligned} &= 4 \int_0^b \int_0^{\frac{a}{b} \sqrt{b^2 - y^2}} dx dy \\ &= 4 \int_0^b \left[x \right]_0^{\frac{a}{b} \sqrt{b^2 - y^2}} dy \\ &= 4 \int_0^b \frac{a}{b} \sqrt{b^2 - y^2} dy \\ &= \frac{4a}{b} \int_0^b \sqrt{b^2 - y^2} dy \\ &= \frac{4a}{b} \left[\frac{y}{2} \sqrt{b^2 - y^2} + \frac{b^2}{2} \sin^{-1} \left(\frac{y}{b} \right) \right]_0^b \\ &= \frac{4a}{b} \left[\frac{b^2}{2} \sin^{-1}(1) \right] \\ &= \frac{4a}{b} \times \frac{b^2}{2} \times \frac{\pi}{2} \\ &= \pi ab \text{ sq. units.} \end{aligned}$$

4) Find the area of the circle with radius 'a'.



Given: Equation of a circle is $x^2 + y^2 = a^2$

$$x^2 = a^2 - y^2$$

$$x = \pm \sqrt{a^2 - y^2}$$

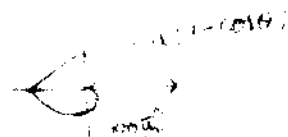
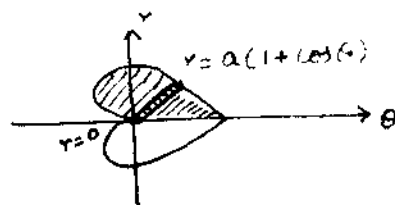
Limits:

x varies from 0 to $\sqrt{a^2 - y^2}$

y varies from 0 to a

$$\begin{aligned}
 \text{Area} &= 4 \times \text{Area of a circle in I quadrant} \\
 &= 4 \int_0^a \int_0^{\sqrt{a^2-y^2}} dx dy \\
 &= 4 \int_0^a [x]_0^{\sqrt{a^2-y^2}} dy \\
 &= 4 \int_0^a \sqrt{a^2-y^2} dy \\
 &= 4 \left[\frac{y}{2} \sqrt{a^2-y^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{y}{a}\right) \right]_0^a \\
 &= 4 \left[\frac{a^2}{2} \sin^{-1}(1) \right] \\
 &= 4 \frac{a^2}{2} \times \frac{\pi}{2} \\
 &= \pi a^2 \text{ sq. units}
 \end{aligned}$$

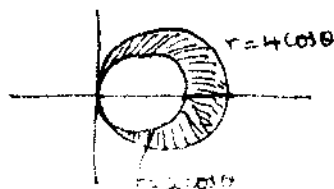
5) Find the area of the cardioid $r = a(1 + \cos \theta)$



Limit: r varies from 0 to $a(1 + \cos \theta)$
 θ varies from 0 to π

$$\begin{aligned}
 \text{Area} &= 2 \times \text{Area of the region in upper half of the plane} \\
 &= 2 \int_0^\pi \int_0^{a(1+\cos \theta)} r dr d\theta \\
 &= 2 \int_0^\pi \left[\frac{r^2}{2} \right]_0^{a(1+\cos \theta)} d\theta \\
 &= 2 \int_0^\pi \frac{a^2(1+\cos \theta)^2}{2} d\theta \\
 &= a^2 \int_0^\pi (1 + \cos^2 \theta + 2 \cos \theta) d\theta \\
 &= a^2 \int_0^\pi \left(1 + \frac{1}{2} + \frac{\cos 2\theta}{2} + 2 \cos \theta \right) d\theta \\
 &= a^2 \left[\frac{3}{2} \theta + \frac{\sin 2\theta}{4} + 2 \sin \theta \right]_0^\pi \\
 &= a^2 \cdot \frac{3}{2} \pi \\
 &= \frac{3\pi a^2}{2} \text{ sq. units}
 \end{aligned}$$

- 6) Find the area of the region outside the inner circle $r = 2\cos\theta$ and inside the outer circle $r = 4\cos\theta$



circle

r varies from $2\cos\theta$ to $4\cos\theta$

θ varies from 0 to $\pi/2$

$$\text{Area} = 2 \int_0^{\pi/2} \int_{r=2\cos\theta}^{4\cos\theta} r dr d\theta$$

$$= 2 \int_0^{\pi/2} \left[\frac{r^2}{2} \right]_{2\cos\theta}^{4\cos\theta} d\theta$$

$$= \int_0^{\pi/2} [(4\cos\theta)^2 - (2\cos\theta)^2] d\theta$$

$$= \int_0^{\pi/2} (16\cos^2\theta - 4\cos^2\theta) d\theta$$

$$= \int_0^{\pi/2} 12\cos^2\theta d\theta$$

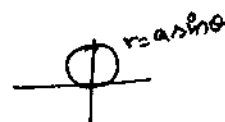
$$= 12 \int_0^{\pi/2} \left(\frac{1}{2} + \frac{\cos 2\theta}{2} \right) d\theta$$

$$= 12 \left[\frac{1}{2}\theta + \frac{\sin 2\theta}{4} \right]_0^{\pi/2}$$

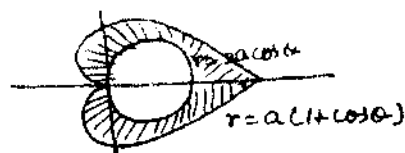
$$= 12 \times \frac{1}{2} \times \frac{\pi}{2}$$

$$= 3\pi \text{ sq. units}$$

Note:



- 7) Find the area of the region outside the circle $r = 2a\cos\theta$ and inside the cardioid $r = a(1+\cos\theta)$



circle: $r = 2a\cos\theta$

$$\text{Area} = 2 \int_0^{\pi/2} \int_0^{2a\cos\theta} r dr d\theta$$

$$= 2 \int_0^{\pi/2} \left[\frac{r^2}{2} \right]_0^{2a\cos\theta} d\theta$$

$$\begin{aligned}
 &= \int_0^{\pi/2} 4a^2 \cos^2 \theta \, d\theta \\
 &= 4a^2 \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta \\
 &= 4a^2 \left[\frac{1}{2} \theta + \frac{\sin 2\theta}{4} \right]_0^{\pi/2} \\
 &= 4a^2 \times \frac{1}{2} \times \frac{\pi}{2} \\
 &= \pi a^2 \text{ sq. units}
 \end{aligned}$$

Cardioid: $r = a(1 + \cos \theta)$

$$\begin{aligned}
 \text{Area} &= 2 \int_0^{\pi} \int_0^{a(1+\cos \theta)} r \, dr \, d\theta \\
 &= 2 \int_0^{\pi} \left[\frac{r^2}{2} \right]_0^{a(1+\cos \theta)} d\theta \\
 &= \int_0^{\pi} a^2 (1 + \cos \theta)^2 d\theta \\
 &= a^2 \int_0^{\pi} [1 + \cos^2 \theta + 2\cos \theta] d\theta \\
 &= a^2 \int_0^{\pi} \left[1 + \frac{1}{2} + \frac{\cos 2\theta}{2} + 2\cos \theta \right] d\theta \\
 &= a^2 \left[\frac{3}{2} \theta + \frac{\sin 2\theta}{4} + 2\sin \theta \right]_0^{\pi} \\
 &= a^2 \times \frac{3}{2} \times \pi \\
 &= \frac{3\pi a^2}{2} \text{ sq. units}
 \end{aligned}$$

$\therefore$ Area of the shaded region = Area of the cardioid - Area of a circle

$$= \frac{3\pi a^2}{2} - \pi a^2$$

$$= \frac{\pi a^2}{2} \text{ sq. units} //$$

* change of order of integration:

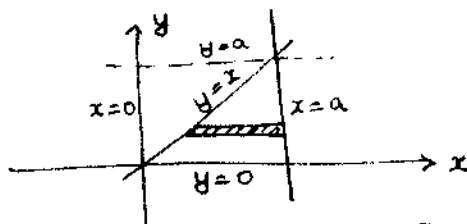
In a double integral, if the limits of integration are constant then the order of integration is immaterial provided the limits of integration are changed accordingly

$$\int_{y=c}^a \int_{x=a}^b f(x,y) dx dy = \int_{x=a}^b \int_{y=c}^d f(x,y) dy dx$$

1) Find the area of integration in $\int_0^a \int_0^x f(x,y) dy dx$

Given: The region of integration bounded by

$$\begin{aligned} y=0 & : y=x \\ x=0 & : x=a \end{aligned} \quad dy dx \rightarrow \text{vertical strip}$$



After changing the order of integration

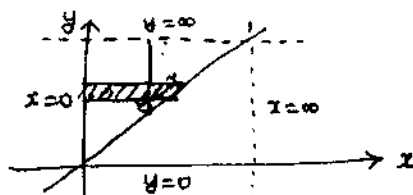
$$\begin{aligned} x & \text{ varies from } y \text{ to } a \\ y & \text{ varies from } 0 \text{ to } a \end{aligned} \quad dx dy \rightarrow \text{Horizontal strip}$$

$$\int_0^a \int_0^x f(x,y) dy dx = \int_{y=0}^a \int_{x=y}^a f(x,y) dx dy$$

2) change of order of the integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$

Given: The region of integration bounded by

$$\begin{aligned} y=x & : y=\infty \\ x=0 & : x=\infty \end{aligned} \quad dy dx \rightarrow \text{vertical strip}$$



After changing the order of the integration

$$\begin{aligned} x & \text{ varies from } 0 \text{ to } y \\ y & \text{ varies from } 0 \text{ to } \infty \end{aligned} \quad dx dy \rightarrow \text{Horizontal strip}$$

$$\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx = \int_{y=0}^\infty \int_{x=0}^y \frac{e^{-y}}{y} dx dy$$

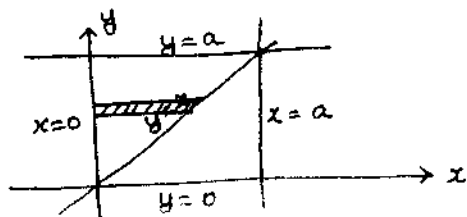
$$\begin{aligned}
 &= \int_0^{\infty} \left[\frac{e^{-y}}{y} x \right]_0^y dy \\
 &= \int_0^{\infty} e^{-y} dy \\
 &= \left[\frac{e^{-y}}{-1} \right]_0^{\infty} \\
 &= [-e^{-\infty} + e^0] \\
 &= 1 // \checkmark
 \end{aligned}$$

- 2) change the order of integration in $\int_0^1 \int_0^y f(x,y) dx dy$
 * change of order of integration in $\int_0^{\infty} \int_0^{\infty} \frac{e^{-y}}{y} dy dx$.

- 3) Evaluate $\int_0^a \int_x^a (x^2 + y^2) dy dx$ by change of order of integration:

Given: The region of integration bounded by

$$\begin{aligned}
 y=x &; y=a \\
 x=0 &; x=a
 \end{aligned}
 \quad dy dx \rightarrow \text{vertical strip}$$



After changing the order of integration

x varies from 0 to y

$dy dx \rightarrow$ Horizontal strip

y varies from 0 to a

$$\begin{aligned}
 \int_0^a \int_x^a (x^2 + y^2) dy dx &= \int_{y=0}^a \int_{x=0}^y (x^2 + y^2) dx dy \\
 &= \int_0^a \left(\frac{x^3}{3} + xy^2 \right) \Big|_0^y dy \\
 &= \int_0^a \left(\frac{y^3}{3} + y^3 \right) dy \\
 &= \left[\frac{y^4}{12} + \frac{y^4}{4} \right]_0^a \\
 &= \frac{a^4}{12} + \frac{a^4}{4} \\
 &= \frac{4a^4}{12} \\
 &= \frac{a^4}{3} \text{ sq. units} //
 \end{aligned}$$

4) Evaluate $\int_0^1 \int_0^{1-x} (x^2 + y^2) dy dx$ by change of order of integration.

5) E

Rewrite the given integral

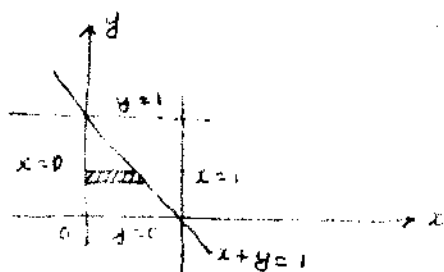
$$\int_0^1 \int_0^{1-x} (x^2 + y^2) dy dx \quad \text{--- (1)}$$

the region of integration bounded by

$$y=0 ; y=1-x \Rightarrow x+y=1$$

$dy dx \rightarrow$ Vertical strip

$$x=0 ; x=1$$



After changing the order of integration

x varies from 0 to $1-y$

$dx dy \rightarrow$ Horizontal strip

y varies from 0 to 1

$$\begin{aligned} \int_0^1 \int_0^{1-x} (x^2 + y^2) dy dx &= \int_0^1 \int_0^{1-y} (x^2 + y^2) dx dy \\ &= \int_0^1 \left(\frac{x^3}{3} + xy^2 \right) \Big|_0^{1-y} dy \\ &= \int_0^1 \left[\frac{(1-y)^3}{3} + y^2(1-y) \right] dy \\ &= \int_0^1 \left[\frac{(1-y)^3}{3} + y^2 - y^3 \right] dy \\ &= \left[-\frac{(1-y)^4}{12} + \frac{y^3}{3} - \frac{y^4}{4} \right]_0^1 \end{aligned}$$

$$= \left\{ 0 + \frac{1}{3} - \frac{1}{4} \right\} - \left\{ -\frac{1}{12} + 0 - 0 \right\}$$

$$= \frac{1}{3} - \frac{1}{4} + \frac{1}{12}$$

$$= \frac{2}{12}$$

$$= \frac{1}{6} \text{ sq. units}$$

5) Evaluate $\int_0^a \int_y^a \frac{x}{\sqrt{x^2+y^2}} dy dx$ by change of order of integration.

Rewrite the given integral

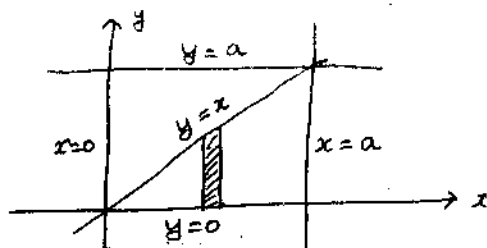
$$\int_0^a \int_y^a \frac{x}{\sqrt{x^2+y^2}} dx dy \quad \text{--- (1)}$$

The region of integration bounded by

$$y=0 \quad ; \quad y=a$$

$dx dy \rightarrow$ Horizontal strip

$$x=y \quad ; \quad x=a$$



$$\therefore \int \frac{dx}{\sqrt{x^2+a^2}} = \log(x + \sqrt{x^2+a^2})$$

After changing the order of integration

x varies from 0 to a

$dy dx \rightarrow$ vertical strip

y varies from 0 to x

$$\begin{aligned} \int_0^a \int_y^a \frac{x}{\sqrt{x^2+y^2}} dx dy &= \int_0^a \int_0^x \frac{x}{\sqrt{x^2+y^2}} dy dx \\ &= \int_0^a x \left[\int_0^x \frac{dy}{\sqrt{x^2+y^2}} \right] dx \\ &= \int_0^a x \left[\log(y + \sqrt{y^2+x^2}) \right]_0^x dx \\ &= \int_0^a x \left[\log(x + \sqrt{x^2+x^2}) - \log x \right] dx \\ &= \int_0^a x \log \left[\frac{(1+\sqrt{2})x}{x} \right] dx \\ &= \log(1+\sqrt{2}) \int_0^a x dx \\ &= \log(1+\sqrt{2}) \left[\frac{x^2}{2} \right]_0^a \\ &= \frac{a^2 \log(1+\sqrt{2})}{2} \quad \text{Ans.} \end{aligned}$$

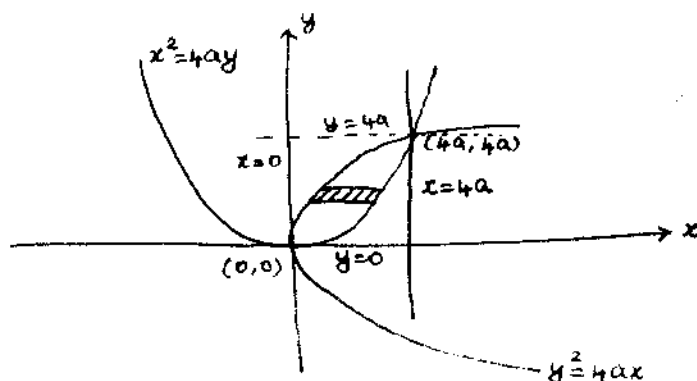
6) Evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xy \, dy \, dx$ or change of order of integration.

7)

Region: The region of integration bounded by

$$y = \frac{x^2}{4a} \Rightarrow x^2 = 4ay \quad \left| \quad x=0 ; x=4a \right.$$

$$y = 2\sqrt{ax} \Rightarrow y^2 = 4ax \quad \left| \quad dy \, dx \rightarrow \text{vertical strip} \right.$$



After changing the order of integration

x varies from $\frac{y^2}{4a}$ to $2\sqrt{ay}$ $dx \, dy \rightarrow$ Horizontal strip

y varies from 0 to $4a$

$$\begin{aligned} \int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xy \, dy \, dx &= \int_0^{4a} \int_{y^2/4a}^{2\sqrt{ay}} xy \, dx \, dy \\ &= \int_0^{4a} \left[\frac{yx^2}{2} \right]_{y^2/4a}^{2\sqrt{ay}} dy \\ &= \frac{1}{2} \int_0^{4a} y \left[4ay - \frac{y^4}{16a^2} \right] dy \\ &= \frac{1}{2} \int_0^{4a} \left(4ay^2 - \frac{y^5}{16a^2} \right) dy \\ &= \frac{1}{2} \left[\frac{4ay^3}{3} - \frac{y^6}{16 \times 6a^2} \right]_0^{4a} \\ &= \frac{1}{2} \left[\frac{(4a)^4}{3} - \frac{(4a)^6}{(4a)^2 \times 6} \right] \\ &= \frac{1}{2} \times 256a^4 \left(\frac{1}{3} - \frac{1}{6} \right) \\ &= \frac{1}{2} \times \overset{64}{\cancel{256}} a^4 \times \frac{1}{6} \\ &= \frac{64}{3} a^4 \text{ sq. units} \end{aligned}$$

7) Evaluate $\int_0^{\infty} \int_0^y y e^{-\frac{y^2}{x}} dx dy$ by change of order

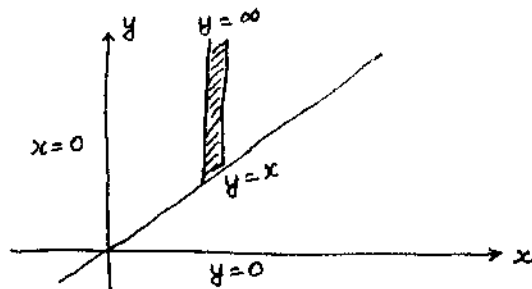
Given:

The region of integration bounded by

$$x=0 ; x=y$$

$dx dy \rightarrow$ horizontal strip

$$y=0 ; y=\infty$$



After changing the order of integration

x varies from 0 to ∞

$dy dx \rightarrow$ vertical strip

y varies from x to ∞

$$\int_0^{\infty} \int_0^y y e^{-\frac{y^2}{x}} dx dy = \int_{x=0}^{\infty} \int_{y=x}^{\infty} y e^{-\frac{y^2}{x}} dy dx$$

$$= \int_0^{\infty} \left[\int_{x^2}^{\infty} e^{-\frac{t}{x}} \frac{dt}{2} \right] dx$$

$$= \frac{1}{2} \int_0^{\infty} \left[\frac{e^{-\frac{t}{x}}}{-\frac{1}{x}} \right]_{x^2}^{\infty} dx$$

$$= \frac{1}{2} \int_0^{\infty} -x [e^{-\infty} - e^{-1}] dx$$

$$= \frac{1}{2} \int_0^{\infty} x e^{-x} dx$$

$$= \frac{1}{2} \left[x \left(\frac{e^{-x}}{-1} \right) - \left(\frac{e^{-x}}{-1} \right) \right]_0^{\infty}$$

$$= \frac{1}{2} \left[-x e^{-x} + e^{-x} \right]_0^{\infty}$$

$$= \frac{1}{2} [-0 + 1]$$

$$= \frac{1}{2} \text{ sq. units}$$

$$\begin{aligned} \text{put } t &= \frac{y^2}{x} \\ dt &= 2y \frac{dy}{x} \\ \therefore y dy &= \frac{dt}{2} \end{aligned}$$

$$\begin{aligned} e^{-\infty} &= 0 \\ e^0 &= 1 \end{aligned}$$

8) Change the order of the integration $\int_0^a \int_y^a \frac{x}{x^2+y^2} dx dy$ and evaluate.

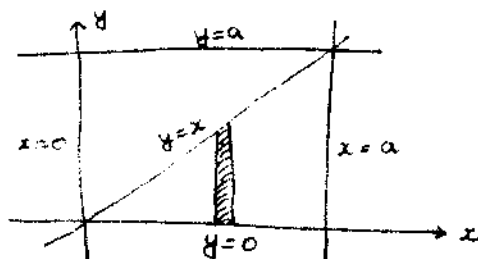
9) c

The region of integration bounded by

$$x=y, \quad x=a$$

$dx dy \rightarrow$ Horizontal strip

$$y=0, \quad y=x$$



After changing the order of integration

$$y=0 : y=x$$

$dy dx \rightarrow$ Vertical strip

$$x=0 : x=a$$

$$\begin{aligned} \int_0^a \int_y^a \frac{x}{x^2+y^2} dx dy &= \int_{x=0}^a \int_{y=0}^x \frac{x}{x^2+y^2} dy dx \\ &= \int_0^a x \left\{ \int_0^x \frac{dy}{x^2+y^2} \right\} dx \\ &= \int_0^a x \left[\frac{1}{x} \tan^{-1} \left(\frac{y}{x} \right) \right]_0^x dx \\ &= \int_0^a \left[\tan^{-1} \left(\frac{x}{x} \right) - \tan^{-1}(0) \right] dx \\ &= \int_0^a \frac{\pi}{4} dx \\ &= \frac{\pi}{4} \int_0^a dx \\ &= \frac{\pi}{4} [x]_0^a \\ &= \frac{\pi a}{4} \text{ sq. units} \end{aligned}$$

and

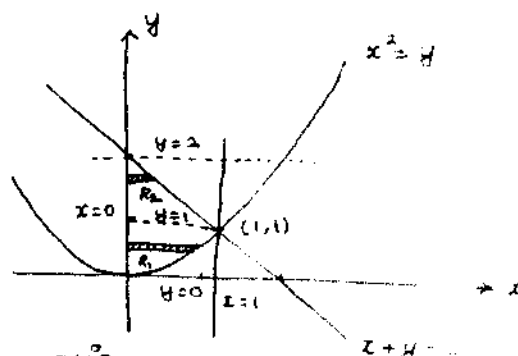
q) change the order of the integration and evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$

The region of integration bounded by

$$\begin{array}{l|l} y^2 = x^2 \text{ --- (1)} & x=0 ; x=1 \quad dy \, dx \rightarrow \text{vertical strip} \\ y = 2-x \text{ --- (2)} & \end{array}$$

Sub (2) in eqn (1)

$$\begin{aligned} 2-x &= x^2 \\ \Rightarrow x^2 + x - 2 &= 0 \\ (x+2)(x-1) &= 0 \\ x &= -2, 1 \\ y &= 4, 1 \end{aligned}$$



After changing the order of integration

Region: R_1

x varies from 0 to $\sqrt{y}$

y varies from 0 to 1

Region: R_2

x varies from 0 to $2-y$

y varies from 1 to 2

$dx \, dy \rightarrow$ horizontal strip

$$\begin{aligned} \int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx &= \iint_{R_1} xy \, dy \, dx + \iint_{R_2} xy \, dx \, dy \\ &= \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy \\ &= \int_0^1 \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy + \int_1^2 \left[\frac{x^2}{2} \right]_0^{2-y} dy \\ &= \frac{1}{2} \int_0^1 y^2 dy + \frac{1}{2} \int_1^2 y(2-y)^2 dy \\ &= \frac{1}{2} \left[\frac{y^3}{3} \right]_0^1 + \frac{1}{2} \int_1^2 y(4+y^2-4y) dy \\ &= \frac{1}{2} \left[\frac{1}{3} - 0 \right] + \frac{1}{2} \left[4 \frac{y^2}{2} + \frac{y^4}{4} - 4 \frac{y^3}{3} \right]_1^2 \\ &= \frac{1}{6} + \frac{1}{2} \left\{ -\left(\frac{4}{2} + \frac{1}{4} - \frac{4}{3} \right) + \left(\frac{16}{2} + \frac{16}{4} - \frac{32}{3} \right) \right\} \\ &= \frac{1}{6} + \frac{1}{2} \left(\frac{5}{12} \right) \\ &= \frac{1}{6} + \frac{5}{24} \\ &= \frac{9}{24} \\ &= \frac{3}{8} \text{ sq. units} \end{aligned}$$

10) change the order of the integration $\int_0^1 \int_y^{2-y} xy \, dx \, dy$ and then evaluate it.

11)

The region of integration bounded by

$$x = y \text{ --- (1)} \quad x = 2 - y \text{ --- (2)} \quad y = 0 \quad ; \quad y = 1 \quad dx \, dy \rightarrow \text{horizontal strip}$$

sub (1) in eqn (2)

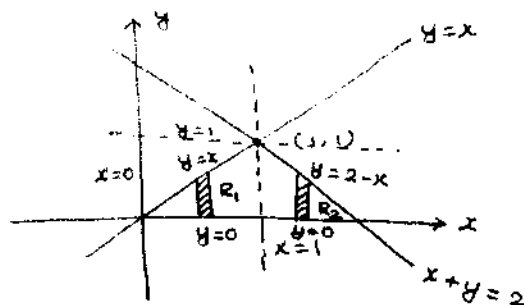
$$x = 2 - y$$

$$y = 1$$

$$y = 0$$

sub $y = 1$ in eqn (1)

$$x = 1$$



the point of intersection (0,0) & (1,1)

After changing the order of integration

Region: R_1

x varies from 0 to 1

y varies from 0 to x

Region: R_2

x varies from 1 to 2

y varies from 0 to $2-x$

$dy \, dx \rightarrow$ vertical strip

$$\begin{aligned} \int_0^1 \int_y^{2-y} xy \, dx \, dy &= \iint_{R_1} xy \, dy \, dx + \iint_{R_2} xy \, dy \, dx \\ &= \int_0^1 \int_0^x xy \, dy \, dx + \int_1^2 \int_0^{2-x} xy \, dy \, dx \\ &= \int_0^1 \left[\frac{xy^2}{2} \right]_0^x dx + \int_1^2 \left[\frac{xy^2}{2} \right]_0^{2-x} dx \\ &= \frac{1}{2} \int_0^1 x^3 dx + \frac{1}{2} \int_1^2 x(2-x)^2 dx \\ &= \frac{1}{2} \left[\frac{x^4}{4} \right]_0^1 + \frac{1}{2} \int_1^2 (4x + x^3 - 4x^2) dx \\ &= \frac{1}{2} \left(\frac{1}{4} \right) + \frac{1}{2} \left[\frac{4x^2}{2} + \frac{x^4}{4} - \frac{4x^3}{3} \right]_1^2 \\ &= \frac{1}{8} + \frac{1}{2} \left\{ \left(\frac{16}{2} + \frac{16}{4} - \frac{32}{3} \right) - \left(\frac{4}{2} + \frac{1}{4} - \frac{4}{3} \right) \right\} \\ &= \frac{1}{8} + \frac{5}{24} \\ &= \frac{8}{24} \\ &= \frac{1}{3} \text{ sq. units} \end{aligned}$$

11). change the order of the integration $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$ then evaluate it.

The region of integration bounded by

$$y = x \text{ --- (1)}$$

$$x = 0$$

$dy dx \rightarrow$ vertical strip

$$y = \sqrt{2-x^2}$$

$$x = 1$$

$$y^2 = 2 - x^2$$

$$\Rightarrow x^2 + y^2 = 2 \text{ --- (2)}$$

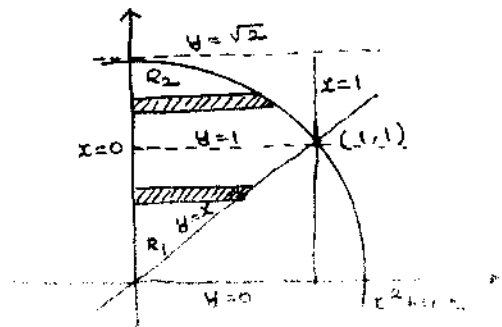
Sub (1) in eqn (2)

$$x^2 + x^2 = 2$$

$$2x^2 = 2$$

$$x^2 = 1$$

$$x = \pm 1$$



Sub in eqn (1) we get $y = \pm 1$

After changing the order of integration

Region: R1

x varies from 0 to y

y varies from 0 to 1

Region: R2

x varies from 0 to $\sqrt{2-y^2}$

y varies from 1 to $\sqrt{2}$

$dx dy \rightarrow$ horizontal strip

$$\begin{aligned} \int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx &= \iint_{R_1} \frac{x}{\sqrt{x^2+y^2}} dx dy + \iint_{R_2} \frac{x}{\sqrt{x^2+y^2}} dx dy \\ &= \int_0^1 \int_0^y \frac{x}{\sqrt{x^2+y^2}} dx dy + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}} dx dy \\ &= \int_0^1 \left[\ln(x^2+y^2) \right]_0^y dy + \int_1^{\sqrt{2}} \left[\ln(x^2+y^2) \right]_0^{\sqrt{2-y^2}} dy \\ &= \int_0^1 (\ln(2y^2) - \ln(y^2)) dy + \int_1^{\sqrt{2}} (\ln(2-y^2) - \ln(y^2)) dy \\ &= \int_0^1 (\sqrt{2} - 1) dy + \int_1^{\sqrt{2}} (\sqrt{2-y^2} - y) dy \\ &= \left[(\sqrt{2}-1)y^2 \right]_0^1 + \left[\sqrt{2}y - y^2 - \frac{y^3}{3} \right]_1^{\sqrt{2}} \\ &= \sqrt{2} - 1 + \left(\sqrt{2} \cdot \sqrt{2} - (\sqrt{2})^2 - \frac{(\sqrt{2})^3}{3} \right) - \left(\sqrt{2} - 1 - \frac{1}{3} \right) \\ &= \sqrt{2} - 1 + 2 - 2 - \frac{2\sqrt{2}}{3} - \sqrt{2} + 1 + \frac{1}{3} \\ &= \frac{1}{3} - \frac{2\sqrt{2}}{3} \end{aligned}$$

* change of order of integration and hence evaluate $\int_1^4 \int_{\sqrt{y}}^2 (x^2 + y^2) dx dy$.

$$\int_1^4 \int_{\sqrt{y}}^2 (x^2 + y^2) dx dy$$

$$\int_0^2 \int_0^{\sqrt{4-y}} x^2 dx dy$$

$$\int_0^3 \int_0^{\sqrt{4-y}} (x+y) dx dy$$

$$\int_0^2 \int_0^x dy dx$$

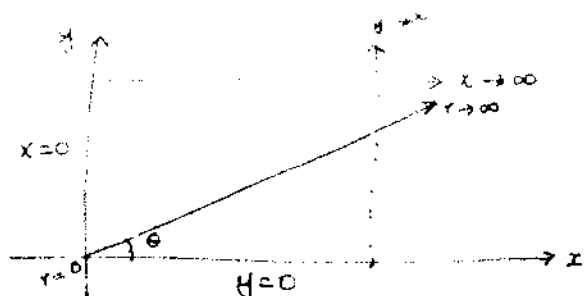
(2) prove that $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \frac{\pi}{4}$ and hence evaluate $\int_0^\infty e^{-x^2} dx$

the region of integration bounded by

$$x=0 \quad ; \quad x=\infty$$

$$y=0 \quad ; \quad y=\infty$$

$dx dy \rightarrow$ horizontal strip



changing to polar co-ordinates

$$x = r \cos \theta, \quad y = r \sin \theta \quad ; \quad dx dy = r dr d\theta \quad ; \quad x^2 + y^2 = r^2$$

r varies from 0 to ∞

θ varies from 0 to $\pi/2$

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^\infty e^{-t} \frac{dt}{2} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \left[\frac{e^{-t}}{-1} \right]_0^\infty d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} (-e^{-\infty} + e^0) d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} d\theta$$

$$= \frac{1}{2} [\theta]_0^{\pi/2}$$

$$= \frac{\pi}{4}$$

put $r^2 = t$

$$2r dr = dt$$

$$r dr = \frac{dt}{2}$$

r	0	∞
t	0	∞

Let $I = \int_0^{\infty} e^{-x^2} dx$ and also $I = \int_0^{\infty} e^{-y^2} dy$

$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy = \frac{\pi}{4}$$

$$\int_0^{\infty} e^{-x^2} dx \int_0^{\infty} e^{-y^2} dy = \frac{\pi}{4}$$

$$I^2 = \frac{\pi}{4}$$

$$I = \frac{\sqrt{\pi}}{2}$$

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

UNIT III

Double Integrals

Type I [Problems on Simple Double Integrals]

- | | |
|---|---|
| 1. $\int_0^1 \int_1^2 (x^2 + y^2) dx dy$ | 2. $\int_4^2 \int_1^2 \frac{dy dx}{(x+y)^2}$ |
| 3. $\int_0^a \int_0^b xy(x - y) dy dx$ | 4. $\int_0^1 \int_1^{x^2} (x^2 + y^2) dy dx$ |
| 5. $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dy dx$ | 6. $\int_0^2 \int_x^{x^2} e^{y/x} dy dx$ |
| 7. $\int_0^a \int_0^{\sqrt{ay}} xy dx dy$ | 8. $\int_0^a \int_0^{\sqrt{ax-x^2}} x^2 dy dx$ |
| 9. $\int_0^a \int_0^{\sqrt{ax-x^2}} x^2 y dx dy$ | 10. $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} xy dy dx$ |
| 11. $\int_0^{\pi/2} \int_0^{\pi/2} \sin(\theta + \phi) d\theta d\phi$ | 12. $\int_{-\pi/2}^{\pi/2} \int_0^{2 \cos \theta} r^2 dr d\theta$ |

❖ Evaluate the following integrals.

- (i) $\iint_R (x^2 + y^2) dx dy$; $R: 0 \leq x \leq a \text{ \& } 0 \leq y \leq a$.
- (ii) $\iint_D (4 - x^2 - y^2) dx dy$; $D: x = 0, x = 1 \text{ \& } y = 0, y = 3/2$
- (iii) $\iint_R (x^2 + y^2) dx dy$; $R: x + y \leq 1$. [+ ve quadrant]
- (iv) $\iint_R (x - y) dx dy$; $R: y = x \text{ \& } y = x^2$.

Type II [To find the Area by using Double integrals]

❖ Find the area from the following curves,

1. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (Apr./May 2018)
2. $4x^2 + 9y^2 = 36 \text{ \& } 2x + 3y = 6$.
3. $y^2 = 4ax \text{ \& } x^2 = 4ay$ (Apr./May 2018, May 2015)
4. $y^2 = 4ax \text{ \& Latus Rectum of Parabola.}$
5. $y^2 = 9x \text{ \& } x^2 = 9y$
6. $y^2 = 4x \text{ \& } x^2 = 4y$

7. Evaluate $\iint (x^2 + y^2) dx dy$ over the area bounded by the curves $y = 4x, x + y = 3; y = 0, y = 2$.

Type III [Transform into Polar Co-ordinates]

❖ Transform into polar co-ordinates and then evaluate the following Double integrals

- $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx.$
- $\int_0^2 \int_0^{\sqrt{2x-x^2}} (x^2 + y^2) dy dx.$
- $\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2 + y^2} dy dx.$
- Evaluate $\iint \frac{x^2 y^2}{(x^2 + y^2)} dx dy$ over the annular region between the circles $x^2 + y^2 = a^2; x^2 + y^2 = b^2 (b > a)$.

Triple Integrals

Type I [Simple Triple Integrals]

- $\int_0^1 \int_0^2 \int_0^3 (x + y + z) dx dy dz.$
- $\int_0^4 \int_0^1 \int_0^1 (x + y + z) dz dy dx.$
- $\int_0^1 \int_0^2 \int_1^2 xyz^2 dz dy dx.$
- $\int_0^a \int_0^b \int_0^c (x + y + z) dz dy dx.$
- $\int_0^1 \int_{y^2}^2 \int_0^{1-x} x dz dx dy.$
- $\int_0^1 \int_0^{1-x} \int_0^{x+y} e^z dx dy dz$ (Model Jan. 2016)
- $\int_0^1 \int_0^y \int_0^{x+y} dx dy dz.$
- $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx.$
- $\int_0^a \int_0^x \int_0^y xyz dz dy dx.$
- $\int_1^3 \int_{1/x}^2 \int_0^{\sqrt{xy}} xyz dz dy dx.$
- $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z dz dx dy.$
- $\int_0^{2\pi} \int_0^{\pi/4} \int_0^a r^2 \sin \theta dr d\theta d\phi.$

❖ **Evaluate the following Triple Integrals**

- (i) $\iiint_R (x + y + z) dx dy dz, R: 1 \leq x \leq 2; 2 \leq y \leq 3; 1 \leq z \leq 3.$
- (ii) $\iiint_R (x - y + z) dx dy dz, R: 1 \leq x \leq 2; 2 \leq y \leq 3; 1 \leq z \leq 3.$
- (iii) $\iiint_D (x + y + z) dx dy dz, D: x = 0, x = 1; y = 0, y = 1; z = 0, z = 1.$
- (iv) $\iiint_V (x + y + z) dx dy dz, V: x + y + z = a; x = 0, y = 0, z = 0.$

Type II [To find the volume by using Triple Integral]

❖ **Find the volume of the following,**

- 1. Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$ (May 2017, Nov 2016)
- 2. Sphere: $x^2 + y^2 + z^2 = a^2$ (Apr./May 2018, May 2016).
- 3. tetrahedron: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ & $x = 0, y = 0, z = 0.$
- 4. $\iiint \frac{dx dy dz}{(x+y+z)^3}$ taken over the volume bounded by $x = 0, y = 0, z = 0$ & $x + y + z = 1.$
- 5. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \sqrt{1-x^2-y^2-z^2} dx dy dz.$

Type III [Transform into cylindrical and spherical co-ordinates]

❖ **Transform into Spherical Co-ordinates**

- 1. $\iiint_R \frac{dx dy dz}{x^2 + y^2 + z^2}, x^2 + y^2 + z^2 = a^2$ (+ve octant).
- 2. Volume of sphere $x^2 + y^2 + z^2 = a^2.$
- 3. $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} xyz dz dy dx.$

❖ **Transform into Cylindrical co-ordinates**

- 1. $\iiint xyz dx dy dz, x^2 + y^2 + z^2 = a^2$ (+ve octant)
- 2. $\iiint (x^2 + y^2 + z^2) dx dy dz, x^2 + y^2 \leq 1$ & $0 \leq z \leq 1$ (sphere)

Change the order of Integration

1. $\int_0^a \int_a^{2\sqrt{ax}} x^2 dy dx.$
2. $\int_0^1 \int_{x^2}^{2-x} xy dy dx.$ (Model May 2016, Jan. 2016, May 2015)
3. $\int_0^a \int_x^a (x^2 + y^2) dy dx.$
4. $\int_0^a \int_y^a \frac{x}{x^2 + y^2} dy dx.$
5. $\int_0^a \int_0^{2\sqrt{ax}} x^2 dy dx.$
6. $\int_0^a \int_y^a \frac{x}{\sqrt{x^2 + y^2}} dy dx.$ (Model Dec. 2017)
7. $\int_0^4 \int_{x^2/4}^{2\sqrt{x}} dy dx.$
8. $\int_0^1 \int_0^x dy dx.$
9. $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xy dy dx.$
10. $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx.$
11. $\int_0^3 \int_1^{\sqrt{4-y}} (x + y) dx dy.$
12. $I = \int_{-a}^a \int_0^{\sqrt{a^2 - x^2}} (x^2 + y^2) dx dy.$